

Education

University of Michigan – Ann Arbor, MS in Information Science (GPA: 3.96/4.0) Aug 2025 - May 2027

University of Michigan – Ann Arbor, BS in Computer Science (GPA: 3.53/4.0) Sept 2020 – Dec 2023

- **Relevant Courses:** **SI 679:** Backend Application Development, **SI 669:** Developing the Mobile Experience, **CSE 593:** Human-Computer Interaction, **SI 644:** Database Application Design, **SI 622:** Needs Assessment and Usability Evaluation, **SI 582:** Interactive Design, **SI 579:** Building Interactive Applications, **EECS 485:** Web Systems, **EECS 482:** Operating Systems, **EECS 442:** Computer Vision

Skills: Python · TypeScript · SQL · C/C++ · Swift · Go · React · Three.js · JSCAD · RealityKit · AWS · Claude Code · Cursor

Experience

SI 544 Graduate Student Instructor · University of Michigan School of Information · Ann Arbor, MI Aug 2026 – Present

- Lead 2 weekly lab sessions for 50+ students in SI 544: Introduction to Statistics and Data Analysis, covering statistical inference, hypothesis testing, data visualization, and multiple linear regression; host office hours and grade coursework.

SWE Intern · Aurora · Pittsburgh, PA Jun 2026 – Aug 2026

- Engineered a web application on top of a dynamic TTL policy CLI tool purging 190PB+ (\$50M+/year) in AWS S3 storage, transforming complex command-line data retention operations lacking visibility into an intuitive visual workflow.
- Integrated the React/TypeScript frontend with Python/Go backend infrastructure via gRPC APIs to seamlessly visualize and optimize custom TTL data lifecycle definitions across Petabyte-scale autonomous driving datasets.
- Enabled non-engineering personnel and leadership to self-serve and independently manage data lifecycle policies.

Research Assistant · University of Michigan Transportation Research Institute (UMTRI) · Ann Arbor, MI May 2025 – Apr 2026

- Online 3D parametric wheelchair - human interaction simulation model humanshape.org/WheelchairTool
 - Implemented modeling methods using different frameworks for 3D human (Three.js) and wheelchair (JSCAD) models, and optimized the interaction between these models, reducing latency per computation by approximately 60%.
- 3D face mask scanning iOS mobile application
 - Developed a 3D, interactive view with custom gesture recognizers using Swift and RealityKit, including a heatmap display of calculated distances between scanned face and mask models using a k-d tree algorithm.

Full Stack Engineer · HELLOMED · Ann Arbor, MI Mar 2024 - May 2025

- Production level medical clinic web application hellomed.com
 - Built and deployed HIPAA-compliant systems for processing sensitive patient data, using presigned URLs, auth tokens, and strict CORS rules; reliably handling 300+ monthly submissions of sensitive patient medical data including images.
 - Drove a 30%+ increase in organic traffic by optimizing SEO via traffic-informed metadata and semantic HTML.

Drill Sergeant · Republic of Korea Army · Imsil, South Korea Feb 2017 - Nov 2018

- Taught and managed 3000+ new recruits in the Recruit Training Battalion by giving them individual combat skills training.

Activities

IT Team Lead · Korean International Student Association (University of Michigan KISA) · Ann Arbor, MI Sep 2022 - Apr 2025

- Founded the IT team and led a team of 7 to proactively build applications addressing the association's IT needs.
- Production level online community web application umichkisa.com
 - Built full-stack servers, clients, and databases for 1k+ daily users, using CloudFront to cut image latency by 95%.
- Real-time food ordering platform umichkisa.com/pocha
 - Developed a live socket notification system for order updates with a fault-tolerant server-side stock algorithm.

President · Korean Scientists and Engineers Association Young Generation (KSEA YG) · Chicago, IL Jan 2020 - Dec 2021

- Served as the first president for the Chicago chapter, representing students across 5+ universities in the area.
- Led events drawing 300+ participants each, and sustained engagement by organizing virtual events during the pandemic.

Projects

Portfolio Website dongsubkim.com

- Designed and implemented a web application using only pure CSS for styling, focusing on design and dynamic transitions.
- Applied a systematic design approach by defining color palettes for dark and light modes, carefully assigning colors to specific UI roles to enhance clarity and visual aspects of user experience, without using any external UI frameworks.

MapReduce Backend Server Simulator (Python)

- Implemented a multi-process and multi-threaded server that concurrently processes high volumes of user text inputs.
- Simulated a fault-tolerant distributed system of managers and workers using TCP and UDP socket clients and servers.